

3.1.22. tautologie? (folosind rezoluția generală)

$$5) \quad ? \quad \models A \vee (B \rightarrow C) \rightarrow ((A \vee B) \rightarrow (A \vee C)) \stackrel{\text{not.}}{=} u$$

$$\neg u \equiv \neg (A \vee (B \rightarrow C) \rightarrow ((A \vee B) \rightarrow (A \vee C)))$$

$$\neg u \equiv \neg (A \vee \neg B \vee C \rightarrow ((\neg A \wedge \neg B) \vee A \vee C))$$

$$\neg u \equiv \neg ((\neg A \wedge B \wedge \neg C) \vee ((\neg A \wedge \neg B) \vee A \vee C))$$

$$\neg u \equiv (A \vee \neg B \vee C) \wedge ((A \vee B) \wedge \neg A \wedge \neg C)$$

$$\neg u = (A \vee \neg B \vee C) \wedge (A \vee B) \wedge \neg A \wedge \neg C \quad \text{FNC cu 4 clauze}$$

$$S = \{ \underset{c_1}{A \vee \neg B \vee C}, \underset{c_2}{A \vee B}, \underset{c_3}{\neg A}, \underset{c_4}{\neg C} \}$$

$$\begin{array}{l} c_2 = A \vee B \\ c_3 = \neg A \end{array} \quad \Bigg|_{\text{res}} \quad B = c_5$$

$$\begin{array}{l} c_1 = A \vee \neg B \vee C \\ c_4 = \neg C \end{array} \quad \Bigg|_{\text{res}} \quad A \vee \neg B = c_6$$

$$\begin{array}{l} c_5 = B \\ c_6 = A \vee \neg B \end{array} \quad \Bigg|_{\text{res}} \quad A = c_7$$

$$\begin{array}{l} c_7 = A \\ c_3 = \neg A \end{array} \quad \Bigg|_{\text{res}} \quad \square \text{ (clauza vidă)} \quad \xrightarrow{\text{TC}} \Rightarrow S \text{-inconsistentă}$$

3.1.24. 5) Strategia eliminării

$$A = \{ p \vee r, \neg q, p \vee q \vee \neg r, \neg p \vee \neg r, q \vee r \}$$

Pas 1: Eliminăm clauze tautologice (nu e cazul)

Pas 2: Eliminarea clauzelor subsumate (nu e cazul)

Pas 3: Eliminarea clauzelor cu literali puri (nu e cazul)

Pas 4: Eliminarea clauzelor unitate $\left\{ \begin{array}{l} \text{eliminăm clauzele care conțin clauza unitate} \\ \text{ștergem negația din clauze} \end{array} \right.$

$\neg q$ - clauză unitate

$$S = \{ p \vee r, p \vee \neg r, \neg p \vee \neg r, r \}$$

r - clauză unitate

$$S = \{ p, \neg p \}$$

p - clauză unitate

$$S = \{ \square \} \text{ inconsistentă}$$

$$S' = \{ p \vee r, p \vee q \vee \neg r, \neg p \vee \neg r, q \vee r \}$$

q - literal pur

$$S' = \{ p \vee r, \neg p \vee \neg r \}$$

$$\begin{array}{l} p \vee r \\ \neg p \vee \neg r \end{array} \quad \Bigg|_{\text{res}} \quad r \vee \neg r \text{ tautologie}$$

$$\Rightarrow S' \text{-consistentă}$$

9.1.25. 5) strategia mulțimii suport

$$\neg p \rightarrow q, (q \rightarrow r) \wedge \neg r \vdash p \wedge \neg r$$

$$u_1 \quad u_2 \quad v$$

$$u_1 \equiv p \vee q \quad (c_1)$$

$$u_2 \equiv (\neg q \vee r) \wedge \neg r$$

$$c_2 \quad c_3$$

$$v \equiv \neg p \vee r \quad (c_4)$$

$$S = \{c_1, c_2, c_3, c_4\}$$

$$Y = \{c_4\} \text{ - mulțimea suport}$$

$$S \setminus Y = \{c_1, c_2, c_3\} \text{ - consistență}$$

$$\text{Res}_r(c_3, c_4) = \neg p \quad (c_5)$$

$$\text{Res}_r(c_1, c_5) = q \quad (c_6)$$

$$\text{Res}_r(c_2, c_6) = r \quad (c_7)$$

$$\text{Res}_r(c_3, c_7) = \square \text{ clauza vidă } \xrightarrow{\text{TCC}} S \text{ - inconsistentă } \Rightarrow \text{ are loc deducția}$$

9.1.19. 5) folosind structura pe nivele

$$p \rightarrow (q \vee r \wedge s), p, \neg r \vdash q \vee r$$

$$u_1 \quad u_2 \quad u_3 \quad v$$

$$u_1: p \rightarrow (q \vee r \wedge s) \equiv \neg p \vee (q \vee r \wedge s) \equiv \neg p \vee q \vee (r \wedge s) \equiv$$

$$\equiv (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee s)$$

$$u_2: p$$

$$u_3: \neg r$$

$$\neg v: \neg (q \vee r) \equiv \neg q \wedge \neg r$$

$$S^0 = S = \{ \neg p \vee q \vee r, \neg p \vee q \vee s, p, \neg r, \neg q \}$$

$$\text{Res}_p(c_1, c_3) = q \vee r = c_6$$

$$\text{Res}_r(c_1, c_4) = \neg p \vee q = c_7$$

$$\text{Res}_q(c_1, c_5) = \neg p \vee r = c_8$$

$$\text{Res}_p(c_2, c_3) = q \vee s = c_9$$

$$\text{Res}_q(c_2, c_5) = \neg p \vee s = c_{10}$$

$$S' = \{ q \vee r, \neg p \vee q, \neg p \vee r, q \vee s, \neg p \vee s \}$$

$$\text{Res}_r(c_6, c_4) = q = c_{11}$$

$$\text{Res}_q(c_6, c_5) = r = c_{12}$$

$$\text{Res}_p(c_7, c_3) = q = c_{11}$$

$$\text{Res}_q(c_7, c_5) = \neg p = c_{13}$$

$$\text{Res}_p(c_8, c_3) = r = c_{12}$$

$$\text{Res}_r(c_8, c_4) = \neg p = c_{13}$$

$$\text{Res}_q(c_9, c_5) = s = c_{14}$$

$$\text{Res}_p(c_{10}, c_3) = \neg = c_{14}$$

$$S^2 = \{ \underset{c_{11}}{q}, \underset{c_{12}}{r}, \underset{c_{13}}{\neg p}, \underset{c_{14}}{\neg} \}$$

$$\text{Res}_2(c_{11}, c_5) = \square \text{ clauza vidă} \Rightarrow \text{are loc consecința logică}$$